

Location Based Service (Research Paper)

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Abstract

The current idea of electric vehicle, is charged to reduce the energy incentive and reduce the cost and time with quick response of location tracking system this goal is achieved through the mobile cellular data service which connected on real time 4G and 5G communication system backend cloud computing is enabled all services that deal with the energy charging system to provide the facility on road real time situation to charge the vehicle on demand and on request using the location based service of cellular system. In this qualitative research method with using the secondary data analysis, data has been collected from reliable sources such as Google Scholars, ACM, MDPI, and IEEE Xplore. Location based service vastly used for mobile based marketing service without any investment, the software prototype model that simulate the model with successful requirement of scalable and achievable result. Mobile edge computing provides the enhance nearest location feature with enhance capability of security feature. Offering a secure protocol which is founded on cryptography to protect the personal information with insubstantial verification. This mechanism is used to secure the unknown user access.

Background & History of Location Based Service

(Song, 2012) Various novel mobile application has been developed due to various technology emergence, various application providers connect map and location service from the cellular device and provide the on demand on request service, such as food delivery service, goods freight service using airline system, customer can easily track and view products from mobile phone. However the various service connected real time using internet 4G system to provide the real time instance service. Existing application using Google map service to find the destination easily, Google API connect on cell phone by using Google services such as Gmail, Chat, Google search engine, that is main component to provide the exact location service which is required by the user. User can share their current information and location using the augmented reality framework runs on smart phone. (Mu-HsingKuo, et al., 2009) Proposed location based service part of mobile communication service which deal with two important factors such as expansion of electric commerce location based service model which provide the interface of particular area that demands the services using internet connection. Method to evaluate the long term and short term cellular services which is mainly component of cellular service application.

(Lammeren, et al., 2009) STEAD approach which is connected with human sensor which is used on location based service to sense the historic facts of landscape. The research finding based on what, where, when, who to use the service using 150 people data that uses the information based location service and sense the system to experiencing the cultural history of dominant mentality group that sense and predict using the location based service. (Hanacde, et al., 2019) Planning for camera surveillance using the multi-source location based on service data. The growing volume of efficient camera planning to configure the video surveillance method to achieve the outcomes of location based monitoring that achieve the target of mobile police, the idea is to predict the interface of camera coverage to predict the footprint, point of interest, and social network data collected from wechat service. The camera service enable the mobile user to obtain the real time monitoring service that recover from the disaster and terrorism activity.

Research Method

In this qualitative research method with using the secondary data analysis, data has been collected form reliable sources such as Google Scholars, ACM, MDPI, and IEEE Xplore.

Location based communication model

Mobile Marketing Using Location Based Service

(C, 2002) Earlier emerging the various application into one platform using the mobile platform that make a great achievement over the mobile user due to the excellent development of communication model with instant sending service. Various computing and telecommunications technologies with the advance development of strong internet speed produced the World Wide Web communication model to achieve the success of mobile web. Mobile web is managing the user needs that control the information service which is execute on anytime and anywhere including the location based service. Location based service vastly used for mobile based marketing service without any investment, the software prototype model that simulate the model with successful requirement of scalable and achievable result. (Yun, et al., 2013) The issue is concerned with privacy, with post adoption behavior of location based service. The survey to collect the data of 234 user which tested on hypothesis the effect of privacy and location service is major issue to control and manage the location, location sharing information may be harmful for the young generation.

An Architecture Pattern To Extend the Interaction Model between Web Services the Location Based Service Context

(Muro-Medrano, 2003) the information is mainly available to customize the service of current internet technology with rapid prototyping model including standard communication protocol, web technology support distributed computing service which interact the mobile user application, the solution to execute the restricted application over the internet using the virtual private network service is also enable the customize sharing service to adding the distributed platform of network traffic that deal with the major achievement of web coordination service.

Secure Augmented Reality Based Recommended System Including Location Based Service

(J.Purnima & M.Vetripriya, 2021) Earlier the recent development of augmented reality based location service that achieve the accurate result of security, privacy, with private network to secure the location service from unknown user. The evolution of android mobile application with smart features of augmented reality based service to propose the location service model using the nearby user service to achieve the goal of privacy concern.

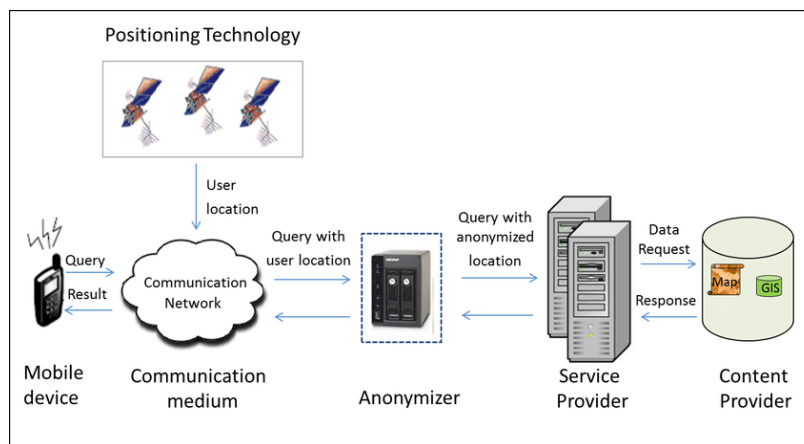


Figure 1: Communication model of mobile technology

Pros & Cons of Location based service

(Bessam A. Adem 1, 2021) The enhance framework of location based service including hybrid model that deal with the framework of transaction service, which is not secure due to other privacy concern issue exist. So banking sector offers privacy service with their own sake to

secure the transactions, the transaction from other site which is not secure due to less security feature. The combine framework of location based service that deal with various vulnerability of network traffic over the network model that does not relate to the technology awareness service. The agent server is introduced to mitigate the schemes of vulnerability to secure the smart network devices for building the hybrid network models with region based location service map.

Delay-aware privacy preserving location based services under spatiotemporal constraints

(Shahid & Pissinou, 2019) Smart services using with smart devices that extract the huge amount of data to control the massive smart devices, the hacker can easily access the cellular data due to delay service and capture the sensitive information from the user, the highlight of research focused on location based hacking system which based on network vulnerability attack that cannot be control due to low security service and GSM delay protocol. KLAP service model which is based on three dimension of delivery requirement of minimum k- location with i-diversity.

Prediction and Anticipation Features-Based Intellectual Assistant in Location Based Service

(Ajay Kr. Gupta (MMM University of Technology, 2021) Introducing assisting technology in healthcare system using the location based service to minimize the control of unknown access. The trajectory data mining method the purpose of achieve the travel route to the user needs, due to overhead cause of precision to compare the results with existing system that control the entire data and privacy system of the cellular system under the privacy act law. (Kanade & Prasad, 2021) Due to novel innovation of augmented reality feature which used to secure the location service location service is portable service to the client which control and administer the orientation of security. The iPhone and android phone gadget based application that improve the geo-information with Google map for getting the information on exact matching algorithm.

Security Issues

Security issues concern on the delivery of the information over the internet service due to delay in packet delivery, hacker might be able to capture the information packet and trace the location of the user which is major security concern on cellular device security.

Mutual Authentication with Wearable Device in Location Based Mobile Edge Computing

(Chin-Ling Chen, pages 575–598) Mature care wearable device which used to achieve the goal of secure communication due to increasing the age the factor of disease has been increase due to less carefulness. Using the data of blood pressure and heartbeat through wearable devices that capture the abnormal condition, the device is able to send the request to the nearest hospital and able to recall the doctor for treatment using the alert messes for nearest location.

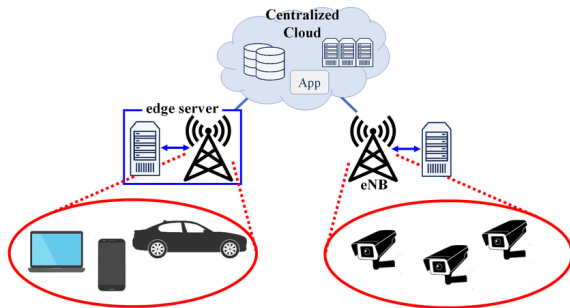


Figure 2: mobile edge computing

Mobile edge computing provides the enhance nearest location feature with enhance capability of security feature. Offering a secure protocol which is founded on cryptography to protect the personal information with insubstantial verification. This mechanism is used to secure the unknown user access.

Data Security in Mobile Cloud Computing

(Rida Qayyum, 2020) Mobile cloud computing concerned with mobile devices which provide the services and information to the cellular user. Cloud computing offer resource concentrating task with less data of cell phone. Mobile cloud computing provide the services to the client on demand service just click to execute the task pattern. The primary concern in the mobile cloud computing to secure the user data with enhance features. (Hayes & FrancescoCappa, 2020) By analyzing the 6 mobile devices which is able to verify the security that commonly collected the mobile data with permission request, the dynamic analysis of cloud computing over the secure feature cover the edge mobile computing under the fog computing concept that cover relay over relay to secure the overall content. The important concern to secure private and public cloud with using the internet service provider is major concern to secure the mobile information.

Current Research (Related Work)

(Ni, et al., 2021) In recent the planning to secure the mobile cloud computing infrastructure over the internet access, the concept of mobile edge computing is working actively to achieve the goal of secure cellular communication with execution of secure location service. The major concern of privacy, security and trust is establish on every second due to reliable communication of cloud system, the service oriented and location based system which respectively works to meet the challenges using the distribution protocol of SOLEK with excellent delivery of information suing the edge computing environment.

SOA (Service Oriented Architecture) Architecture of Location Based Service of Mobile

(Pop, et al., 2021) The architecture is considering due to complete delivery of information with reliable and secure features, the data is transfer for using wireless communication channel by using the REST and SOAP web service using mobile client software application of android and iPhone user. The concept of industry 4.0 is revolutionized with smart communication of mobile phone with reliable packet delivery due to enhance 4G and 5G service feature.

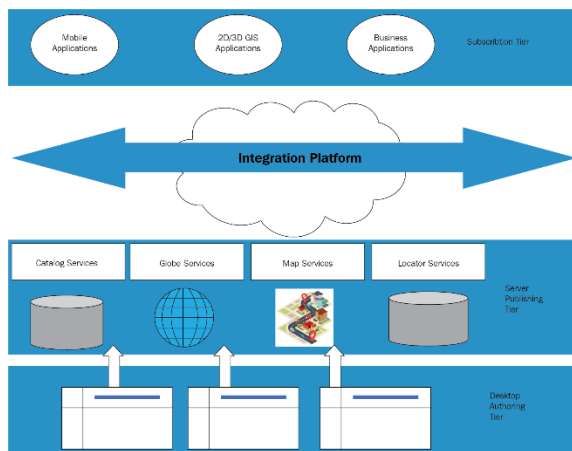


Figure 3: Service Oriented architecture using mobile platform

Emerging Technologies using the Mobile application

(A. F. Del Carpio and Y. P. Vera, 2021) rapid development of mobile application merge the all technology into one umbrella, Google S2 map location service provides various API to the mobile cloud computing environment to draw the attention of location with secure socket layer of SSL the merging of virtual reality and augmented reality using the smart IOT system which is

very useful mechanism to achieve the goal of secure communication. The background of decision oriented software architecture is being a part of smart phone which takes decision on request and analyzed the request according to the given requirement.

Result & Analysis

Result & analysis of location based service based on qualitative data analysis to analyze the case study of context aware service and electric vehicle charging system based on location. Case studies analyzed and explored using the method of qualitative content analysis method. The following case studies has been carried out in this research.

1. Context Aware Computing
2. Electric Vehicle Charging System

Case Study in Location based service

(Barkhuus, 2003) Context aware computing always proposal undertaking the people location which track from the mobile data, various studies offer to secure the user information in private cloud there are two types of services such as location based service and location tracking service which dealt to track the user location with context aware services modeling. The experiment case studies which analyze the location and privacy concern of the people community over the edge computing, location tracking services is more secure as compare to the location based service.

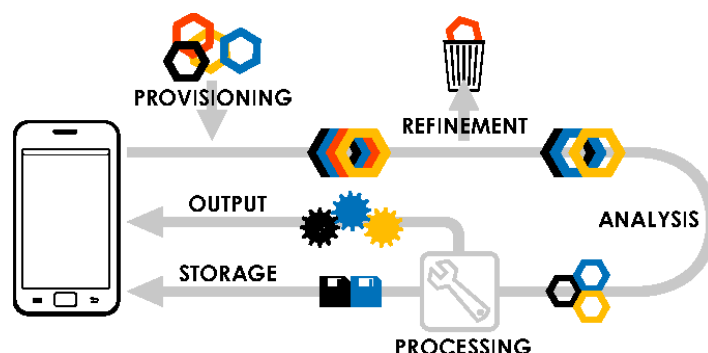


Figure 4: Context aware computing

Electric Vehicle Charging System Using the Location Based Incentive to Reducing the Energy

(Sung-WonPark, 2021) the modern concept of electric vehicle, is charged to reduce the energy incentive and reduce the cost and time with quick response of location tracking system this goal

is achieved through the mobile cellular data service which is connected on real time 4G and 5G communication system. Backend cloud computing is enabled all services that deal with the energy charging system to provide the facility on road real time situation to charge the vehicle on demand and on request. The case study captured from Jeju Island in Korea which proposes a software response model on real situation tackling service.

Discussion & Limitation

(Gupta & Shani, et al., 553-561) The recommended system for location based service which addresses the communication process to manage the security challenges to provide the private end to end encryption service to the mobile user, the spectrum service of location based which organizes the way of communication system. In these days the easiest method to identify and explore the new things which are based on favorite restaurant, shop, bus ticket, and online payment procedure. It is under the one umbrella service, today E-commerce merges with mobile commerce communication due to emerging technologies of edge computing. Offers two location based services such as collaborative and content based filtering technique that personalizes the user experience on demand delivery of information with various location history pattern algorithm that matches with user requirement and produces the exact result with pattern matching algorithm, the apriori algorithm founded on exact user behavior pattern with various time zone and location type. The second algorithm deals with generalized location recommender with collaborative sifting technique, to solve the service pattern by using the k-means clustering algorithm also deployed.

K-Nearest Neighbor Privacy Protection Query for distributed storage in location based service

(Yibing Li, 2021) Fast growth of global putting system with various mobile system that relatively merge all the application into one platform with advancement of modern algorithm method. People request location based service from cellular service provider using the mobile devices platform. Modern method to deal the complexity of difficult problem in which the problem exists to solve location service with modern mechanism such as k-nearest neighbor query based cryptography method instead of using the z order curve and Hilbert curve to store the information. The algorithm uses the mechanism of homomorphism to keep the query in

confidential which support the storage service and location based service more secure in order to meet the challenges.

Future Research & Conclusion

In this research paper the investigation of mobile service based location is highlight the important thing to concluded that the future research required to secure the public cloud with dealing of cellular service provide, future research required to improve the load balancing method over the cloud and edge computing to secure the location with region based service. Various algorithm used to secure the location and secure the information such as k-nearest neighbor algorithm, clustering algorithm, is better method to secure the personal information of the user. The suggested scheme for location based service which respect the communication process to manage the security challenges to provide the private end to end encryption service to the mobile user, the spectrum service of location based which organized the way of communication system.

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